

Patterns, prediction and data leakage

Does the model recognize biology, or merely recognize the donor?

By the end, you should be able to...

Distinguish clustering from supervised prediction.

Split repeated observations by the intended independent unit.

Read a confusion matrix and compare against a negative control.

Reconnect with the biology

A feature is a measured input; a label is the outcome being predicted.
Rows from the same donor can share a signature.

Think silently -> explain to a neighbour -> check the measurement.

Two different questions

Explore patterns

Predict labels

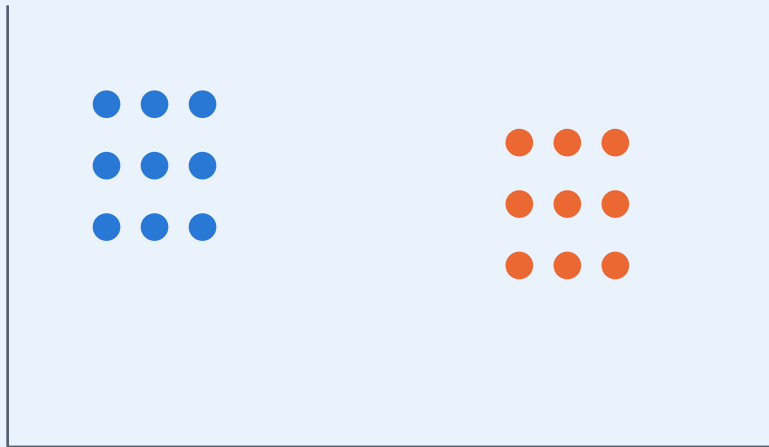
Clustering groups observations using similarity without supplied class labels.

Classification learns a mapping from features to known labels.

A visual cluster is not automatically a cell type or a disease mechanism.

A projection is a view, not the complete data

Illustrative 2D projection



A displayed group is a hypothesis, not a diagnosis.

PCA summarizes directions of variation.

UMAP emphasizes a neighborhood representation and depends on parameters.

Distances between plotted islands do not automatically measure biological separation.

Define the prediction target first

01

Will the model be applied to another observation from a known donor?

>

02

Or to a new donor, study, hospital or time period?

>

03

The test split must represent the intended use.

Repeated donors can leak across a row split

Train	Test
Donor A rows	Donor A rows
Donor B rows	Donor B rows
Wrong for a new-donor claim	
Train	Test
Donors A + B	Donors C + D

Near-duplicate observations can identify their donor.

A row-level split may put the same donor in train and test.

Keep all observations from a donor together for new-donor evaluation.

Preprocessing can leak too

01

Split first. Fit scaling and feature selection on training data.

>

02

Use the fitted transform on validation or test data.

>

03

Do not choose a method by repeatedly inspecting the final test score.

Four counts explain a classifier

Truth	Pred +	Pred -
Positive	TP = 8	FN = 2
Negative	FP = 1	TN = 9

Sensitivity $8/10 = 0.80$

Specificity $9/10 = 0.90$

True positive and false negative describe actual positives.

False positive and true negative describe actual negatives.

Sensitivity and specificity use different denominators.

Accuracy needs context

95% accuracy

Can coexist with zero sensitivity

If 95 of 100 samples are negative, always predicting negative gives 95% accuracy.

But sensitivity is zero.

Report the class balance, independent sample count and uncertainty.

A shuffled-label control respects the design

01

For repeated donors, shuffle labels at donor level.

>

02

Keep all observations of a donor assigned the same shuffled label.

>

03

Repeat the control; chance-level performance varies in small test sets.

Worked example: read a confusion matrix

TP=8, FN=2, TN=9, FP=1; the test set contains 20 independent individuals.

Predict the result and one limitation before calculating.

Work through the reasoning

$$\text{Sensitivity} = 8 / (8 + 2) = 0.80.$$

$$\text{Specificity} = 9 / (9 + 1) = 0.90.$$

$$\text{Accuracy} = (8 + 9) / 20 = 0.85; \text{ balanced accuracy} = (0.80 + 0.90) / 2 = 0.85.$$

Pause: make a defensible claim

A dataset contains six rows for each of 12 donors. A random row split gives 100% accuracy. What split would test performance on new donors?

First answer alone. Then compare the evidence, not just the conclusion.

A defensible response

Hold out whole donors so no donor is shared across train and test, then report donor-level performance and the small independent test size.

Today's practical

W09 / Project D2: compare a deliberately leaky row split with an honest donor split, calculate confusion counts and run a donor-level permutation control.

Minutes	Action
0-10	Predict + set up
10-30	Run + inspect + check
30-40	Explain + compare
40-45	Save + submit / exit

Five terms to use precisely

- 1 feature
- 2 label
- 3 train/test
- 4 leakage
- 5 sensitivity

Exit ticket

What did we actually measure or calculate?

What is one assumption or limitation of that result?

What check would you perform before sharing it?

Go further only after the core is clear

Optional: nested cross-validation and calibration; no neural-network training is required.

Not required for the core grade. No advanced model training is expected.

Read the evidence, not only the slide

S11 scikit-learn: Common pitfalls and recommended practices.

S13 AnnData documentation: annotated data matrices.